Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow A = ...,\ B = ...$	M1	1.1b
	Either $A = 2$ or $B = -1$	A1	1.1b
	$\frac{3}{(2x-1)(x+1)} = \frac{2}{2x-1} - \frac{1}{x+1}$	A1	1.1b
		(3)	
(b)	$\int \frac{1}{V}\,dV = \int \frac{3}{(2t-1)(t+1)}\,dt$	B1	1.1a
	$\int \frac{2}{2t-1} - \frac{1}{t+1}\,dt = ...\ln(2t-1) - ...\ln(t+1)(+c)$	M1	3.1a
	$\ln V = \ln(2t-1) - \ln(t+1)(+c)$	A1ft	1.1b
	Substitutes $t = 2, V = 3 \Rightarrow c = (\ln 3)$	M1	3.4
	$\ln V = \ln(2t-1) - \ln(t+1) + \ln 3$ $V = \frac{3(2t-1)}{(t+1)}$ *	A1*	2.1
		(5)	

	(b) Alternative separation of variables:		
	$\int \frac{1}{3V}\,dV = \int \frac{1}{(2t-1)(t+1)}\,dt$	B1	1.1a
	$\frac{1}{3}\int \frac{2}{2t-1} - \frac{1}{t+1}\,dt = ...\ln(2t-1) - ...\ln(t+1)(+c)$	M1	3.1a
	$\frac{1}{3}\ln 3V = \frac{1}{3}\ln(2t-1) - \frac{1}{3}\ln(t+1)(+c)$	A1ft	1.1b
	Substitutes $t = 2, V = 3 \Rightarrow c = \left(\frac{1}{3}\ln 3\right)$	M1	3.4
	$\frac{1}{3}\ln V = \frac{1}{3}\ln(2t-1) - \frac{1}{3}\ln(t+1) + \frac{1}{3}\ln 3$ $V = \frac{3(2t-1)}{(t+1)}$ *	A1*	2.1
		(5)	
(c)	(i) 30 (minutes)	B1	3.2a
	(ii) 6 (m^3)	B1	3.4
		(2)	
			(10 marks)
Notes:			

(a)

M1: Correct method of partial fractions leading to values for their A and B

E.g. substitution: $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = A(x+1) + B(2x-1) \Rightarrow A = ..., B = ...$

Or compare coefficients $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = x(A+2B) + A - B \Rightarrow A = ..., B = ...$

Note that $\frac{3}{(2x-1)(x+1)} = \frac{A}{2x-1} + \frac{B}{x+1} \Rightarrow 3 = A(2x-1) + B(x+1) \Rightarrow A = ..., B = ...$ scores M0

A1: Correct value for "A" or "B"

A1: Correct partial fractions not just values for "A" and "B". $\frac{2}{2x-1}-\frac{1}{x+1}$ or e.g. $\frac{2}{2x-1}+\frac{-1}{x+1}$

Must be seen as **fractions** but if not stated here, allow if the correct fractions appear later.

(b)

B1: Separates variables $\int \frac{1}{V}\mathrm{d}V = \int \frac{3}{(2t-1)(t+1)}\mathrm{d}t$. May be implied by later work.

Condone omission of the integral signs but the dV and dt must be in the correct positions if awarding this mark in isolation but they may be implied by subsequent work.

M1: Correct attempt at integration of the partial fractions.

Look for $...\ln(2t-1)+...\ln(t+1)$ where ... are constants.

Condone missing brackets around the $(2t-1)$ and/or the $(t+1)$ for this mark

A1ft: Fully correct equation following through their A and B **only**.

No requirement for $+c$ here.

The brackets around the $(2t-1)$ and/or the $(t+1)$ must be seen or implied for this mark

M1: Attempts to find "c" or e.g. "$\ln k$" using $t=2,\ V=3$ following an attempt at integration.

Condone poor algebra as long as $t=2,\ V=3$ is used to find a value of their constant.

Note that the constant may be found immediately after integrating or e.g. after the ln's have been combined.

A1*: Correct processing leading to the given answer $V=\frac{3(2t-1)}{(t+1)}$

Alternative:

B1: Separates variables $\int \frac{1}{3V}\mathrm{d}V = \int \frac{1}{(2t-1)(t+1)}\mathrm{d}t$. May be implied by later work.

Condone omission of the integral signs but the dV and dt must be in the correct positions if awarding this mark in isolation but they may be implied by subsequent work.

M1: Correct attempt at integration of the partial fractions.

Look for $...\ln(2t-1)+...\ln(t+1)$ where ... are constants.

Condone missing brackets around the $(2t-1)$ and/or the $(t+1)$ for this mark

A1ft: Fully correct equation following through their A and B **only**.

No requirement for $+c$ here.

The brackets around the $(2t-1)$ and/or the $(t+1)$ must be seen or implied for this mark

M1: Attempts to find "c" or e.g. "$\ln k$" using $t=2,\ V=3$ following an attempt at integration.

Condone poor algebra as long as $t=2,\ V=3$ is used to find a value of their constant.

Note that the constant may be found immediately after integrating or e.g. after the ln's have been combined.

A1*: Correct processing leading to the given answer $V=\frac{3(2t-1)}{(t+1)}$

(Note the working may look like this:

$$\frac{1}{3}\ln 3V=\frac{1}{3}\ln(2t-1)-\frac{1}{3}\ln(t+1)+c,\ \frac{1}{3}\ln 9=\frac{1}{3}\ln(3)-\frac{1}{3}\ln 3+c,\ c=\frac{1}{3}\ln 9$$

$$\ln 3V=\ln\frac{9(2t-1)}{(t+1)}\Rightarrow 3V=\frac{9(2t-1)}{(t+1)}\Rightarrow V=\frac{3(2t-1)}{(t+1)}*)$$

Note that B0M1A1M1A1 is not possible in (b) as the B1 must be implied if all the other marks have been awarded.

Note also that some candidates may use different variables in (b) e.g.

$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{3y}{(2x-1)(x+1)}\Rightarrow\int\frac{1}{y}\mathrm{d}y=\int\frac{3}{(2x-1)(x+1)}\mathrm{d}x$ etc. In such cases you should award marks for equivalent work but they must revert to the given variables at the end to score the final mark.

Also if e.g. a "t" becomes an "x" within their working but is recovered allow full marks.

(c)

B1: Deduces 30 minutes. Units not required so just look for 30 but allow equivalents e.g. ½ an hour.

If units are given they must be correct so do not allow e.g. 30 hours.

B1: Deduces 6 m^3. Units not required so just look for 6. Condone $V<6$ or $V\leq 6$

If units are given they must be correct so do not allow e.g. 6 m.

Q2.

Question	Scheme	Marks	AOs
	$C: y = x\ln x$; l is a normal to C at $P(\mathrm{e}, \mathrm{e})$ Let x_A be the x-coordinate of where l cuts the x-axis		
	$\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x + x\left(\frac{1}{x}\right) \quad \{= 1 + \ln x\}$	M1 A1	2.1 1.1b
	$x = \mathrm{e},\ m_T = 2 \Rightarrow m_N = -\frac{1}{2} \Rightarrow y - \mathrm{e} = -\frac{1}{2}(x - \mathrm{e})$ $y = 0 \Rightarrow -\mathrm{e} = -\frac{1}{2}(x - \mathrm{e}) \Rightarrow x = \ldots$	M1	3.1a
	l meets x-axis at $x = 3\mathrm{e}$ (allow $x = 2\mathrm{e} + \mathrm{e}\ln\mathrm{e}$)	A1	1.1b
	{Areas:} **either** $\int_1^{\mathrm{e}} x\ln x\,\mathrm{d}x = [\ \ldots\ ]_1^{\mathrm{e}} = \ldots$ **or** $\frac{1}{2}((\text{their } x_A) - \mathrm{e})\mathrm{e}$	M1	2.1
	$\left\{\int x\ln x\,\mathrm{d}x = \right\}\ \frac{1}{2}x^2\ln x - \int \frac{1}{x}\cdot\left(\frac{x^2}{2}\right)\{\mathrm{d}x\}$	M1	2.1
	$\left\{= \frac{1}{2}x^2\ln x - \int \frac{1}{2}x\,\{\mathrm{d}x\}\right\} = \frac{1}{2}x^2\ln x - \frac{1}{4}x^2$	dM1 A1	1.1b 1.1b
	$\text{Area}(R_1) = \int_1^{\mathrm{e}} x\ln x\,\mathrm{d}x = [\ \ldots\ ]_1^{\mathrm{e}} = \ldots$; $\text{Area}(R_2) = \frac{1}{2}((\text{their } x_A) - \mathrm{e})\mathrm{e}$ and so, $\text{Area}(R) = \text{Area}(R_1) + \text{Area}(R_2) \quad \{= \tfrac{1}{4}\mathrm{e}^2 + \tfrac{1}{4} + \mathrm{e}^2\}$	M1	3.1a
	$\text{Area}(R) = \tfrac{5}{4}\mathrm{e}^2 + \tfrac{1}{4}$	A1	1.1b
		(10)	

Notes for Question	
M1:	Differentiates by using the product rule to give $\ln x + x(\text{their } g'(x))$, where $g(x) = \ln x$
A1:	Correct differentiation of $y = x\ln x$, which can be un-simplified or simplified
M1:	Complete strategy to find the x coordinate where their normal to C at $P(\mathrm{e}, \mathrm{e})$ meets the x-axis i.e. Sets $y = 0$ in $y - \mathrm{e} = m_N(x - \mathrm{e})$ to find $x = \ldots$
Note:	m_T is found by using calculus and $m_N \neq m_T$
A1:	l meets x-axis at $x = 3\mathrm{e}$, allowing un-simplified values for x such as $x = 2\mathrm{e} + \mathrm{e}\ln\mathrm{e}$
Note:	Allow $x =$ awrt 8.15
M1:	Scored for either • Area under curve $= \int_1^{\mathrm{e}} x\ln x\,\mathrm{d}x = [\ \ldots\ ]_1^{\mathrm{e}} = \ldots$, with limits of e and 1 and some attempt to substitute these and subtract • or Area under line $= \frac{1}{2}((\text{their } x_A) - \mathrm{e})\mathrm{e}$, with a valid attempt to find x_A
M1:	Integration by parts the correct way around to give $Ax^2\ln x - \int B\left(\frac{x^2}{x}\right)\{\mathrm{d}x\}$; $A \neq 0$, $B > 0$
dM1:	**dependent on the previous M mark** Integrates the second term to give $\pm\lambda x^2$; $\lambda \neq 0$
A1:	$\frac{1}{2}x^2\ln x - \frac{1}{4}x^2$
M1:	Complete strategy of finding the area of R by finding the sum of two key areas. See scheme.
A1:	$\frac{5}{4}\mathrm{e}^2 + \frac{1}{4}$
Note:	$\text{Area}(R_2)$ can also be found by integrating the line l between limits of e and their x_A i.e. $\text{Area}(R_2) = \int_{\mathrm{e}}^{\text{their } x_A}\left(-\frac{1}{2}x + \frac{3}{2}\mathrm{e}\right)\mathrm{d}x = [\ \ldots\ ]_{\mathrm{e}}^{\text{their } x_A} = \ldots$
Note:	**<u>Calculator approach with no algebra, differentiation or integration seen:</u>** • Finding l cuts through the x-axis at awrt 8.15 is 2nd M1 2nd A1 • Finding area between curve and the x-axis between $x = 1$ and $x = \mathrm{e}$ to give awrt 2.10 is 3rd M1 • Using the above information (must be seen) to apply $\text{Area}(R) = 2.0972\ldots + 7.3890\ldots = 9.4862\ldots$ is final M1 Therefore, a maximum of 4 marks out of the 10 available.

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	The asymptote is found where $2x - q = 0$ Hence $q = 4$	B1	This mark is given for explaining that the asymptote at $x = 2$ is a solution of $2x - q = 0$
	$y = \frac{p - 3x}{(2x-4)(x+3)}$ $\frac{1}{2} = \frac{p-9}{(6-4)(3+3)}$	M1	This mark is given for substituting $x = 3$, $y = \frac{1}{2}$ (and $q = 4$)

(b)	$\frac{15-3x}{(2x-4)(x+3)} = \frac{A}{(2x-4)} + \frac{B}{(x+3)}$	M1	This mark is given for a method to use partial fractions
	$= \frac{1.8}{(2x-4)} - \frac{2.4}{(x+3)}$	M1	This mark is given for finding values for A and B
	$= \frac{0.9}{(x-2)} - \frac{2.4}{(x+3)}$	A1	This mark is given for a fully simplified expression
	$I = \int \frac{15-3x}{(2x-4)(x+3)}\,dx$ $= m \ln(2x-4) + n \ln(x+3)$	M1	This mark is given for a method to integrate to find the area of R
	$= 0.9 \ln(2x-4) + 2.4 \ln(x+3)$	A1	This mark is given for a correct expression for the area of R
	Area $R = \left[0.9\ln(2x-4) - 2.4\ln(x+3)\right]_3^5$	M1	This mark is given for deducing an expression for the area of R ($y = 0$ when $x = 5$)
	$= [0.9 \ln 6 - 2.4 \ln 8] - [0.9 \ln 2 - 2.4 \ln 6]$ $= [0.9 \ln 6 + 2.4 \ln 6] - [7.2 \ln 2 + 0.9 \ln 2]$ $= 3.3 \ln 6 - 8.1 \ln 2$ $= 3.3 \ln 3 + 3.3 \ln 2 - 8.1 \ln 2$	M1	This mark is given for a method to find the exact area of R
	$= 3.3 \ln 3 - 4.8 \ln 2$	A1	This mark is given for a correct value of the area of R with $a = 3.3$ and $b = 4.8$

Q4.

Question	Scheme	Marks	AOs
(a)	Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$	B1	1.1a
	Substitutes either $P=0$ or $P=\frac{11}{2}$ into $1 = A(11-2P)+BP \Rightarrow A \text{ or } B$	M1	1.1b
	$\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$	A1	1.1b
		(3)	
(b)	Separates the variables $\int \frac{22}{P(11-2P)} \mathrm{d}P = \int 1 \,\mathrm{d}t$	B1	3.1a
	Uses (a) and attempts to integrate $\int \frac{2}{P} + \frac{4}{(11-2P)} \mathrm{d}P = t + c$	M1	1.1b
	$2\ln P - 2\ln(11-2P) = t + c$	A1	1.1b
	Substitutes $t=0, P=1 \Rightarrow t=0, P=1 \Rightarrow c = (-2\ln 9)$	M1	3.1a
	Substitutes $P = 2 \Rightarrow t = 2\ln 2 + 2\ln 9 - 2\ln 7$	M1	3.1a
	Time $= 1.89$ years	A1	3.2a
		(6)	

(c)	Uses ln laws $2\ln P - 2\ln(11-2P) = t - 2\ln 9$ $\Rightarrow \ln\left(\frac{9P}{11-2P}\right) = \frac{1}{2}t$	M1	2.1
	Makes 'P' the subject $\Rightarrow \left(\frac{9P}{11-2P}\right) = e^{\frac{1}{2}t}$ $\Rightarrow 9P = (11-2P)e^{\frac{1}{2}t}$ $\Rightarrow P = f\left(e^{\frac{1}{2}t}\right)$ or $\Rightarrow P = f\left(e^{-\frac{1}{2}t}\right)$	M1	2.1
	$\Rightarrow P = \frac{11}{2+9e^{-\frac{1}{2}t}} \Rightarrow A = 11, B = 2, C = 9$	A1	1.1b
		(3)	
			(12 marks)

Notes:

(a)

B1: Sets $\dfrac{1}{P(11-2P)} = \dfrac{A}{P} + \dfrac{B}{(11-2P)}$

M1: Substitutes $P=0$ or $P=\dfrac{11}{2}$ into $1 = A(11-2P) + BP \Rightarrow A$ *or* B

Alternatively compares terms to set up and solve two simultaneous equations in A and B

A1: $\dfrac{1}{P(11-2P)} = \dfrac{1/11}{P} + \dfrac{2/11}{(11-2P)}$ or equivalent $\dfrac{1}{P(11-2P)} = \dfrac{1}{11P} + \dfrac{2}{11(11-2P)}$

Note: The correct answer with no working scores all three marks.

(b)

B1: Separates the variables to reach $\displaystyle\int \frac{22}{P(11-2P)}\,\mathrm{d}P = \int 1\,\mathrm{d}t$ or equivalent

M1: Uses part (a) and $\displaystyle\int \frac{A}{P} + \frac{B}{(11-2P)}\mathrm{d}P = A\ln P \pm C\ln(11-2P)$

A1: Integrates both sides to form a correct equation including a 'c' Eg $2\ln P - 2\ln(11-2P) = t + c$

M1: Substitutes $t=0$ and $P=1$ to find c

M1: Substitutes $P=2$ to find t. This is dependent upon having scored both previous M's

A1: Time = 1.89 years

(c)

M1: Uses correct log laws to move from $2\ln P - 2\ln(11-2P) = t + c$ to $\ln\left(\dfrac{P}{11-2P}\right) = \dfrac{1}{2}t + d$ for their numerical 'c'

M1: Uses a correct method to get P in terms of $e^{\frac{1}{2}t}$

This can be achieved from $\ln\left(\dfrac{P}{11-2P}\right) = \dfrac{1}{2}t + d \Rightarrow \dfrac{P}{11-2P} = e^{\frac{1}{2}t+d}$ followed by cross multiplication and collection of terms in P (See scheme)

Alternatively uses a correct method to get P in terms of $e^{-\frac{1}{2}t}$ For example $\dfrac{P}{11-2P} = e^{\frac{1}{2}t+d} \Rightarrow \dfrac{11-2P}{P} = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \dfrac{11}{P} - 2 = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \dfrac{11}{P} = 2 + e^{-\left(\frac{1}{2}t+d\right)}$ followed by division

A1: Achieves the correct answer in the form required. $P = \dfrac{11}{2+9e^{-\frac{1}{2}t}} \Rightarrow A = 11, B = 2, C = 9$ oe

Q5.

Question	Scheme	Marks	AOs
(a)	$u=1+\sqrt{x}\Rightarrow x=(u-1)^2\Rightarrow\frac{\mathrm{d}x}{\mathrm{d}u}=2(u-1)$ or $u=1+\sqrt{x}\Rightarrow\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2}x^{-\frac{1}{2}}$	B1	1.1b
	$\int\frac{x}{1+\sqrt{x}}\mathrm{d}x=\int\frac{(u-1)^2}{u}2(u-1)\mathrm{d}u$ or $\int\frac{x}{1+\sqrt{x}}\mathrm{d}x=\int\frac{x}{u}\times2x^{\frac{1}{2}}\mathrm{d}u=\int\frac{2x^{\frac{3}{2}}}{u}\mathrm{d}u=\int\frac{2(u-1)^3}{u}\mathrm{d}u$	M1	2.1
	$\int_0^{16}\frac{x}{1+\sqrt{x}}\mathrm{d}x=\int_1^5\frac{2(u-1)^3}{u}\mathrm{d}u$	A1	1.1b
		(3)	
(b)	$2\int\frac{u^3-3u^2+3u-1}{u}\mathrm{d}u=2\int\left(u^2-3u+3-\frac{1}{u}\right)\mathrm{d}u=\ldots$	M1	3.1a
	$=(2)\left[\frac{u^3}{3}-\frac{3u^2}{2}+3u-\ln u\right]$	A1	1.1b
	$=2\left[\frac{5^3}{3}-\frac{3(5)^2}{2}+3(5)-\ln 5-\left(\frac{1}{3}-\frac{3}{2}+3-\ln 1\right)\right]$	dM1	2.1
	$=\frac{104}{3}-2\ln 5$	A1	1.1b
		(4)	
			(7 marks)

Notes
(a) B1: Correct expression for $\frac{\mathrm{d}x}{\mathrm{d}u}$ or $\frac{\mathrm{d}u}{\mathrm{d}x}$ (or u') or dx in terms of du or du in terms of dx M1: Complete method using the given substitution. This needs to be a correct method for their $\frac{\mathrm{d}x}{\mathrm{d}u}$ or $\frac{\mathrm{d}u}{\mathrm{d}x}$ leading to an integral in terms of u only (ignore any limits if present) so for each case you need to see: $\frac{\mathrm{d}x}{\mathrm{d}u}=\mathrm{f}(u) \to \int \frac{x}{1+\sqrt{x}}\mathrm{d}x = \int \frac{(u-1)^2}{u}\mathrm{f}(u)\mathrm{d}u$ $\frac{\mathrm{d}u}{\mathrm{d}x}=\mathrm{g}(x) \to \int \frac{x}{1+\sqrt{x}}\mathrm{d}x = \int \frac{x}{u}\times\frac{\mathrm{d}u}{\mathrm{g}(x)} = \int \mathrm{h}(u)\mathrm{d}u$. In this case you can condone slips with coefficients e.g. allow $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2}x^{-\frac{1}{2}} \to \int \frac{x}{1+\sqrt{x}}\mathrm{d}x = \int \frac{x}{u}\times\frac{x^{\frac{1}{2}}}{2}\mathrm{d}u = \int \mathrm{h}(u)\mathrm{d}u$ but not $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2}x^{-\frac{1}{2}} \to \int \frac{x}{1+\sqrt{x}}\mathrm{d}x = \int \frac{x}{u}\times\frac{x^{-\frac{1}{2}}}{2}\mathrm{d}u = \int \mathrm{h}(u)\mathrm{d}u$ A1: All correct with correct limits and no errors. The "du" must be present but may have been omitted along the way but it must have been seen at least once before the final answer. The limits can be seen as part of the integral or stated separately. (b) M1: Realises the requirement to cube the bracket and divide through by u and makes progress with the integration to obtain at least 3 terms of the required form e.g. 3 from ku^3, ku^2, ku, $k\ln u$ A1: Correct integration. This mark can be scored with the "2" still outside the integral or even if it has been omitted. But if the "2" has been combined with the integrand, the integration must be correct. dM1: Completes the process by applying their "changed" limits and subtracts the right way round **Depends on the first method mark.** A1: Cao (Allow equivalents for $\frac{104}{3}$ e.g. $\frac{208}{6}$)

Q6.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\mathrm{d}h = -2(4-u)\,\mathrm{d}u$	B1	This mark is given for finding an expression for dh
	$\int \frac{\mathrm{d}h}{4-\sqrt{h}} = \int \frac{-2(4-u)\,\mathrm{d}u}{4-\sqrt{h}}$	M1	This mark is given for substituting $u = 4 - \sqrt{h}$ into the integral
	$= \int -\frac{8}{u} + 2\,\mathrm{d}u$	M1	This mark is given for a method to find a simplified version of the integral
	$-8 \ln u + 2u + c$ $= -8 \ln(4-\sqrt{h}) + 2(4-\sqrt{h}) + c$	M1	This mark is given for integrating with respect to u to produce an expression in terms of h
		A1	This mark is given for a correct expression for the integral
	$= -8 \ln(4-\sqrt{h}) - 2\sqrt{h} + k$	A1	This mark is given for a full proof to arrive at the answer as shown (appreciating that $k = c + 8$)
(b)	$\frac{\mathrm{d}h}{\mathrm{d}t} = 0 \Rightarrow 4 - \sqrt{h} = 0$	M1	This mark is given for a setting $\frac{\mathrm{d}h}{\mathrm{d}t} = 0$
	$0 < h < 16$	A1	This mark is given for deducing the range of the heights of the trees for this model

(c)	$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{t^{0.25}(4-\sqrt{h})}{20} \Rightarrow \frac{\mathrm{d}h}{(4-\sqrt{h})} = \frac{t^{0.25}\mathrm{d}t}{20}$	B1	This mark is given for separating the variables
	$-8 \ln(4-\sqrt{h}) - 2\sqrt{h} + k = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to integrate both sides of the equation
		A1	This mark is given for integrating both sides of the equation correctly
	When $t = 0$ and $h = 1$, $-8 \ln 3 - 2 + k = 0$ $k = 2 + 8 \ln 3$	M1	This mark is given for substituting values of $t = 0$ and $h = 1$ to find a value for k
	When $h = 12$, $-8 \ln(4-\sqrt{12}) - 2\sqrt{12} + 2 + 8 \ln 3 = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to find a value for t by substituting $h = 12$ into the equation
	$t^{1.25} = 221.2795 \Rightarrow t = \sqrt[1.25]{221.2795}$	M1	This mark is given for simplifying to find an expression for t
	$t = 75.2$ years	A1	This mark is given for correctly finding the time the tree would take to reach a height of 12 metres
			(Total 15 marks)

Q7.

Question	Scheme	Marks	AOs
(a)	Attempts $y.\frac{\mathrm{d}x}{\mathrm{d}t}=(2\sin 2t+3\sin t)\times 16\sin t\cos t$ and uses $\sin 2t=2\sin t\cos t$	M1	2.1
	Correct expanded integrand. Usually for one of $(R)=\int \underline{\underline{48\sin^2 t\cos t+16\sin^2 2t}}\,\mathrm{d}t$ $(R)=\int \underline{\underline{48\sin^2 t\cos t+64\sin^2 t\cos^2 t}}\,\mathrm{d}t$ $(R)=\int \underline{\underline{24\sin 2t\sin t+16\sin^2 2t}}\,\mathrm{d}t$	A1	1.1b
	Attempts to use $\cos 4t=1-2\sin^2 2t=\left(1-8\sin^2 t\cos^2 t\right)$	M1	1.1b
	$R=\int_0^a 8-8\cos 4t+48\sin^2 t\cos t\,\mathrm{d}t$ *	A1*	2.1
	Deduces $a=\frac{\pi}{4}$	B1	2.2a
		(5)	
(b)	$\int 8-8\cos 4t+48\sin^2 t\cos t\,\mathrm{d}t=8t-2\sin 4t+16\sin^3 t$	M1 A1	2.1 1.1b
	$\left[8t-2\sin 4t+16\sin^3 t\right]_0^{\frac{\pi}{4}}=2\pi+4\sqrt{2}$	M1 A1	2.1 1.1b
		(4)	
			(9 marks)
Notes:			

(a) Condone work in another variable, say $\theta \leftrightarrow t$ if used consistently for the first 3 marks

M1: For the key step in attempting $y.\frac{\mathrm{d}x}{\mathrm{d}t}=(2\sin 2t+3\sin t)\times 16\sin t\cos t$ with an attempt to use $\sin 2t=2\sin t\cos t$ Condone slips in finding $\frac{\mathrm{d}x}{\mathrm{d}t}$ but it must be of the form $k\sin t\cos t$

E.g. I $\quad y.\frac{\mathrm{d}x}{\mathrm{d}t}=(2\sin 2t+3\sin t)\times k\sin t\cos t=(4\sin t\cos t+3\sin t)\times k\sin t\cos t$

E.g. II $\quad y.\frac{\mathrm{d}x}{\mathrm{d}t}=(2\sin 2t+3\sin t)\times k\sin t\cos t=(2\sin 2t+3\sin t)\times\frac{k}{2}\sin 2t$

A1: A correct (expanded) integrand in t. Don't be concerned by the absence of $\int$ or $\mathrm{d}t$ or limits

$$(R)=\int \underline{\underline{48\sin^2 t\cos t+16\sin^2 2t}}\,\mathrm{d}t \quad \text{or} \quad (R)=\int \underline{\underline{48\sin^2 t\cos t+64\sin^2 t\cos^2 t}}\,\mathrm{d}t$$

but watch for other correct versions such as $(R)=\int \underline{\underline{24\sin 2t\sin t+16\sin^2 2t}}\,\mathrm{d}t$

M1: Attempts to use $\cos 4t = \pm 1 \pm 2\sin^2 2t$ to get the integrand in the correct form.

If they have the form $P\sin^2 2t$ it is acceptable to write $P\sin^2 2t = \frac{P}{2}(\pm 1 \pm \cos 4t)$

If they have the form $Q\sin^2 t\cos^2 t$ sight and use of $\sin 2t$ and/or $\cos 2t$ will usually be seen first.

There are many ways to do this, below is such an example

$$Q\sin^2 t\cos^2 t = Q\left(\frac{1-\cos 2t}{2}\right)\left(\frac{1+\cos 2t}{2}\right) = Q\left(\frac{1-\cos^2 2t}{4}\right) = Q\left(\frac{1}{4} - \frac{\cos^2 2t}{4}\right) = Q\left(\frac{1}{4} - \frac{1+\cos 4t}{8}\right)$$

Allow candidates to start with the given answer and work backwards using the same rules. So expect to see $\cos 4t = \pm 1 \pm 2 \times \sin^2 2t$ or $\cos 4t = \pm 2 \times \cos^2 2t \pm 1$ before double angle identities for $\sin 2t$ or $\cos 2t$ are used.

A1*: Proceeds to the given answer with correct working. The order of the terms is not important.
Ignore limits for this mark. The integration sign and the dt must be seen on their final answer.
If they have worked backwards there must be a concluding statement to the effect that they know that they have shown it. The integration sign and the dt must also be seen

E.g. Reaches $\int 48\sin^2 t\cos t + 64\sin^2 t\cos^2 t \,\mathrm{d}t$

Answer is
$$\int 8 - 8\cos 4t + 48\sin^2 t\cos t \,\mathrm{d}t$$
$$= \int 8 - 8\left(1 - 2\sin^2 2t\right) + 48\sin^2 t\cos t \,\mathrm{d}t$$
$$= \int 16\sin^2 2t + 48\sin^2 t\cos t \,\mathrm{d}t$$
$$= \int 64\sin^2 t\cos^2 t + 48\sin^2 t\cos t \,\mathrm{d}t$$
which is the same, ✓

B1: Deduces $a = \frac{\pi}{4}$. It may be awarded from the upper limit and can be awarded from (b)

(b)

M1: For the key process in using a correct approach to integrating the trigonometric terms.
May be done separately.
There may be lots of intermediate steps (e.g. let $u = \sin t$).
There are other more complicated methods so look carefully at what they are doing.

$\int 8 - 8\cos 4t + 48\sin^2 t\cos t \,\mathrm{d}t = ... \pm P\sin 4t \pm Q\sin^3 t$ where P and Q are constants

A1: $\int 8 - 8\cos 4t + 48\sin^2 t\cos t \,\mathrm{d}t = 8t - 2\sin 4t + 16\sin^3 t (+c)$

If they have written $16\sin^3 t$ as $16\sin t^3$ only award if further work implies a correct answer.
Similarly, $8t$ may be written as $8x$. Award if further work implies $8t$, e.g. substituting in their limits.
Do not penalise this sort of slip at all, these are intermediate answers.

M1: Uses the limits their a and 0 where $a = \frac{\pi}{6}, \frac{\pi}{4}$ or $\frac{\pi}{3}$ in an expression of the form $kt \pm P\sin 4t \pm Q\sin^3 t$ leading to an exact answer. Ignore evidence at lower limit as terms are 0

A1: CSO $2\pi + 4\sqrt{2}$ or exact **simplified** equivalent such as $2\pi + \frac{8}{\sqrt{2}}$ or $2\pi + \sqrt{32}$.

Be aware that $\int_0^{\frac{\pi}{4}} 8 - 8\cos 4t + 48\sin^2 t\cos t \,\mathrm{d}t = 8t + \underset{=}{\lambda}\sin 4t + 16\sin^3 t (+c)$ would lead to the correct answer but would only score M1 A0 M1 A0

Q8.

Question	Scheme	Marks	AOs
(a)	Uses or implies $h = 0.5$	B1	1.1b
	For correct form of the trapezium rule =	M1	1.1b
	$\frac{0.5}{2}\{3+2.2958+2(2.3041+1.9242+1.9089)\}=4.393$	A1	1.1b
		(3)	
(b)	Any valid statement reason, for example • Increase the number of strips • Decrease the width of the strips • Use more trapezia	B1	2.4
		(1)	
(c)	For integration by parts on $\int x^2 \ln x \, dx$	M1	2.1
	$=\frac{x^3}{3}\ln x-\int \frac{x^2}{3}\,dx$	A1	1.1b
	$\int -2x+5\,dx=-x^2+5x \quad (+c)$	B1	1.1b
	All integration attempted and limits used Area of $S=\int_1^3 \frac{x^2 \ln x}{3}-2x+5\,dx=\left[\frac{x^3}{9}\ln x-\frac{x^3}{27}-x^2+5x\right]_{x=1}^{x=3}$	M1	2.1
	Uses correct ln laws, simplifies and writes in required form	M1	2.1
	Area of $S=\frac{28}{27}+\ln 27 \quad (a=28, b=27, c=27)$	A1	1.1b
		(6)	
			(10 marks)

Notes:	
(a)	
B1:	States or uses the strip width $h = 0.5$. This can be implied by the sight of $\frac{0.5}{2}\{...\}$ in the trapezium rule
M1:	For the correct form of the bracket in the trapezium rule. Must be y values rather than x values $\{\text{first } y \text{ value} + \text{last } y \text{ value} + 2 \times (\text{sum of other } y \text{ values})\}$
A1:	4.393
(b)	
B1:	See scheme
(c)	
M1:	Uses integration by parts the right way around. Look for $\int x^2 \ln x \, dx = Ax^3 \ln x - \int Bx^2 \, dx$
A1:	$\frac{x^3}{3}\ln x - \int \frac{x^2}{3} \, dx$
B1:	Integrates the $-2x+5$ term correctly $= -x^2 + 5x$
M1:	All integration completed and limits used
M1:	Simplifies using ln law(s) to a form $\frac{a}{b} + \ln c$
A1:	Correct answer only $\frac{28}{27} + \ln 27$

Q9.

Question	Scheme for Substitution		Marks	AOs
	Chooses a suitable method for $\int_0^2 2x\sqrt{x+2}\,\mathrm{d}x$ Award for • Using a valid substitution $u =$..., changing the terms to u's • integrating and using appropriate limts .		M1	3.1a
	Substitution $u=\sqrt{x+2} \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=2u$ oe	Substitution $u=x+2 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=1$ oe	B1	1.1b
	$\int 2x\sqrt{x+2}\,\mathrm{d}x$ $=\int A(u^2\pm 2)u^2\mathrm{d}u$	$\int 2x\sqrt{x+2}\,\mathrm{d}x$ $=\int A(u\pm 2)\sqrt{u}\mathrm{d}u$	M1	1.1b
	$=Pu^5 \pm Qu^3$	$=Su^{\frac{5}{2}} \pm Tu^{\frac{3}{2}}$	dM1	2.1
	$=\frac{4}{5}u^5-\frac{8}{3}u^3$	$=\frac{4}{5}u^{\frac{5}{2}}-\frac{8}{3}u^{\frac{3}{2}}$	A1	1.1b
	Uses limits 2 and $\sqrt{2}$ the correct way around	Uses limits 4 and 2 the correct way around	ddM1	1.1b
	$=\frac{32}{15}\left(2+\sqrt{2}\right)$ *		A1*	2.1
			(7)	
				(7 marks)

M1: For attempting to integrate using substitution. Look for

- terms and limits changed to u's. Condone slips and errors/omissions on changing $\mathrm{d}x \to \mathrm{d}u$
- attempted multiplication of terms and raising of at least one power of u by one. Condone slips
- Use of at least the top correct limit. For instance if they go back to x's the limit is 2

B1: For substitution it is for giving the substitution and stating a correct $\frac{\mathrm{d}x}{\mathrm{d}u}$

Eg, $u=\sqrt{x+2} \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=2u$ or equivalent such as $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2\sqrt{x+2}}$

M1: It is for attempting to get all aspects of the integral in terms of 'u'.
All terms must be attempted including the $\mathrm{d}x$. You are only condoning slips on signs and coefficients
dM1: It is for using a correct method of expanding and integrating each term (seen at least once) . It is dependent upon the previous M
A1: Correct answer in x or u See scheme
ddM1: Dependent upon the previous M, it is for using the correct limits for their integral, **the correct way around**

A1*: Proceeds correctly to $=\frac{32}{15}\left(2+\sqrt{2}\right)$. **Note that this is a given answer**

There must be at one least correct intermediate line between $\left[\frac{4}{5}u^5-\frac{8}{3}u^3\right]_{\sqrt{2}}^{2}$ and $\frac{32}{15}\left(2+\sqrt{2}\right)$

Question Alt	Scheme for by parts	Marks	AOs
	Chooses a suitable method for $\int_0^2 2x\sqrt{x+2}\,dx$ Award for • using by parts the correct way around • and using limits	M1	3.1a
	$\int\left(\sqrt{x+2}\right)dx = \frac{2}{3}(x+2)^{\frac{3}{2}}$	B1	1.1b
	$\int 2x\sqrt{x+2}\,dx = Ax(x+2)^{\frac{3}{2}} - \int B(x+2)^{\frac{3}{2}}(dx)$	M1	1.1b
	$= Ax(x+2)^{\frac{3}{2}} - C(x+2)^{\frac{5}{2}}$	dM1	2.1
	$= \frac{4}{3}x(x+2)^{\frac{3}{2}} - \frac{8}{15}(x+2)^{\frac{5}{2}}$	A1	1.1b
	Uses limits 2 and 0 the correct way around	ddM1	1.1b
	$= \frac{32}{15}\left(2+\sqrt{2}\right)$	A1*	2.1
		(7)	

M1: For attempting using by parts to solve It is a problem- solving mark and all elements do not have to be correct.

- the formula applied the correct way around. You may condone incorrect attempts at integrating $\sqrt{x+2}$ for this problem solving mark
- further integration, again, this may not be correct, and the use of at least the top limit of 2

B1: For $\int\left(\sqrt{x+2}\right)dx = \frac{2}{3}(x+2)^{\frac{3}{2}}$ oe May be awarded $\int_0^2 2x\sqrt{x+2}\,dx \to x^2 \times \frac{2(x+2)^{\frac{3}{2}}}{3}$

M1: For integration by parts the right way around. Award for $Ax(x+2)^{\frac{3}{2}} - \int B(x+2)^{\frac{3}{2}}(dx)$

dM1: For integrating a second time. Award for $Ax(x+2)^{\frac{3}{2}} - C(x+2)^{\frac{5}{2}}$

A1: $\frac{4}{3}x(x+2)^{\frac{3}{2}} - \frac{8}{15}(x+2)^{\frac{5}{2}}$ which may be un simplified

ddM1: Dependent upon the previous M, it is for using the limits 2 and 0 the **correct way around**

A1*: Proceeds to $= \frac{32}{15}\left(2+\sqrt{2}\right)$. **Note that this is a given answer.**

At least one correct intermediate line must be seen. (See substitution). You would condone missing dx's